



Cyclic behavior of concrete shear wall with CFRP grid-steel reinforcement

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ABSTRACT

Carbon fiber reinforced polymer (CFRP) grid possesses not only the characteristic of FRP material, but the property of grid form with good integrity and bond behavior. It has been applied in structural reinforcement, and recently in new constructions. In this study, thirteen pieces of concrete shear walls, including two reinforced concrete (RC) specimens and eleven CFRP grid specimens with aspect ratios from 1.0 to 2.2, were tested under reversed loading. A new reinforcement configuration with CFRP grid was proposed in specimen design for partial and fully replacing steel bar. Parameters of reinforcement configuration, reinforcement ratio, and aspect ratio were considered in this investigation of cyclic behavior. The mechanical property of CFRP grid shear wall was revealed based on the evaluation of cyclic behavior. A comprehensive perspective of analysis from shear failure to flexural failure was compared and summarized. In general, the experimental results showed that specimens with CFRP grid performed higher load and displacement capacity and lower residual deformation. CFRP grid specimen showed smaller width of main cracks, surrounded by a large number of hairlike cracks, indicating a better stress distribution.

1. Introduction

Shear wall is one of the widely used lateral load systems for high-rise building structures to resist the actions of wind and earthquake [1]. Reinforced concrete (RC) shear wall is the most common type, with sufficient stiffness and lateral load capacity [2]. Under the effect of aspect ratio, reinforcement ratio, and axial compression ratio, the failure mode of RC wall includes shear failure, flexural failure, and flexural-shear failure, characterized by both shear and flexural response. In structural design, the ductile flexural failure of slender shear walls under the earthquake is expected [3]. However, squat walls are inevitable in some special situations, such as structures with large opening space at the bottom. The whole action will be dominated by shear response, resulting in brittle shear failure [4–6]. Moreover, although well-detailed slender walls possess excellent ductility and energy dissipation, post-earthquake rehabilitation will be a difficult issue. In the coastal area, once the cracks pass through the concrete cover, corrosion would happen in reinforcement, which will undermine the structure.

Fiber reinforced polymer (FRP) material has been applied in construction for years. This material would become a promising alternative reinforcement to achieve rehabilitation and avoid corrosion in

structures, due to the properties of light weight, high strength, linear elasticity, and resistance to corrosion [7–9]. Many scholars focus on the shear wall reinforced with FRP bars. Mohamed et al. [10] concluded that detailed glass-FRP (GFRP) reinforced walls could meet the expected requirements of flexural capacity, drift ratio, and energy dissipation. Arafa et al. [11] found that in GFRP-bar squat walls, horizontal web reinforcement could increase the ultimate load capacity under the domination of diagonal tension. Both horizontal and vertical web reinforcements were essential for crack recovery and concrete contribution enhancement. Ghazizadeh et al. [12] carried on an experimental and analytical study on damage-resistant RC low-rise walls with hybrid GFRP-Steel reinforcement and steel fibers. Specimen with steel fiber-reinforced concrete and FRP bars presented self-centering capacity and damage-resistant performance. Zhao et al. [13] found carbon-FRP (CFRP) reinforced walls achieved expected lateral strength, post-yield stiffness, acceptable lateral drift, and excellent self-centering behavior, although the level of energy dissipation was lower than the RC wall. As mentioned above, FRP bars effectively improve the structural performance of concrete shear wall, as well as avoid corrosion. However, the weak bond behavior and the reduction of strength at the bending point in stirrups are the disadvantages, which still need to be improved.

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CFRP grid, with the same characteristics as FRP bar, has its unique properties due to the structure of bidirectionally distributed fibers [14]. The well-connected transverse and longitudinal fibers of CFRP grid enhance not only the bond behavior between the grid and concrete but the integrity of the whole member [15], especially for large-area plane members (slab and wall). Recently, CFRP grids have been applied in new construction members such as concrete beam, column, and slab [16–26] due to its promising properties. Research on concrete shear wall reinforced with CFRP grid has just started. Yamakawa et al. [27] investigated the elasto-plastic behavior of concrete shear walls reinforced with CFRP grids. Specimens, with fully replaced CFRP grids, showed rapid load-carrying degradation, low energy dissipation, and ductility. High load capacity and full hysteretic loop were achieved as only part of steel reinforcements were replaced with CFRP grids. Huang et al. [28] conducted six concrete shear walls with CFRP grids to investigate the shear behavior of specimens. It found that specimens with CFRP grids showed better shear performance, including larger lateral displacement, load capacity, and lower residual deformation. Song et al. [29] conducted an experimental investigation on six slender shear walls with hybrid CFRP grid and steel reinforcement. The result concluded that CFRP grids enhanced the load-carrying capacity and energy dissipation, reduced the residual deformation. It recommended the reinforcement configuration and hybrid proportion for CFRP grid shear walls.

In general, Huang et al. and Song et al. conducted the research on CFRP grid shear wall from different perspectives, shear behavior and flexural behavior, respectively. However, it is not enough to investigate the cyclic behavior of CFRP grid shear wall separately. On one hand, the shear and flexural responses occur simultaneously during the load, as well as the proportion of two responses will change or even can be ignored with the change of aspect ratio or reinforcement ratio. On the other, the effect of parameter changes on the mechanical property, furthermore on the failure mode is of significance to be figured out, while it appears both in shear and flexural analysis. Thus, a combined analysis needs to be set up to compare the mechanical property and failure characteristics of shear wall changing from shear failure to flexural failure. The evaluation of cyclic performance and the in-depth insight of failure at different failure modes need to be summarized and revealed. Moreover, it will provide a valuable reference for further fundamental research and application of CFRP grid shear wall.

This paper aimed to provide a comprehensive perspective to experimentally investigate the cyclic behavior of concrete shear wall with

CFRP grid-steel reinforcement under quasi-static reversed loading. Thirteen pieces of concrete shear walls were tested to compare and summarize mechanical property and failure characteristics at different failure modes. Cyclic performance evaluations including crack propagation, hysteretic behavior, stiffness degradation, ductility, energy dissipation, residual deformation, and strain development were thoroughly discussed and compared in different failure modes.

2. Experimental program

2.1. Test specimens

Thirteen concrete shear walls (two RC and eleven CFRP grid specimens) were constructed and tested under cyclic loading. The variables included (1) reinforcement configuration, (2) aspect ratio, and (3) vertical and horizontal reinforcement ratios. The specimens were divided into two groups (Groups A and B) according to different dimensions and reinforcement configurations to meet the requirements of the expected failure mode. In Group A, the failure was expected to be dominated by shear response, while in Group B, flexure-dominated failure was expected.

Fig. 1 illustrates the test program and the specimen nomenclature. Fig. 2 shows the reinforcement details of the specimen. In Group A, six specimens shared the same cross section (1000 mm × 100 mm), but different heights (1000 mm and 1400 mm). Similarly, in Group B, seven specimens were in the same cross section (900 mm × 150 mm), but different in height (1400 mm and 2000 mm). The name of the specimen was composed of the group, aspect ratio, and reinforcement configuration, which could be found in Fig. 1. For instance, specimen A1.0-SS belonged to Group A, with the aspect ratio of 1.0 and reinforcement configuration of two vertical steel webs (SS).

Specimen A1.0-SS was the RC shear wall in Group A as a reference. Specimens A1.0-CC and A1.0-HC1 were designed to keep the reinforcement ratio constant, with different configurations. In specimen A1.0-CC, two pieces of CFRP grids were arranged vertically connecting by tie bars, while in specimen A1.0-HC1, CFRP grids were arranged horizontally and fixed with vertical steel bars by steel wire. These three specimens were designed to investigate the effect of reinforcement configuration. In specimen A1.0-HC2, the horizontal reinforcement ratio was half of that of specimen A1.0-HC1. To investigate the effect of aspect ratio, the heights of specimens A1.4-CC and A1.4-HC2 increased

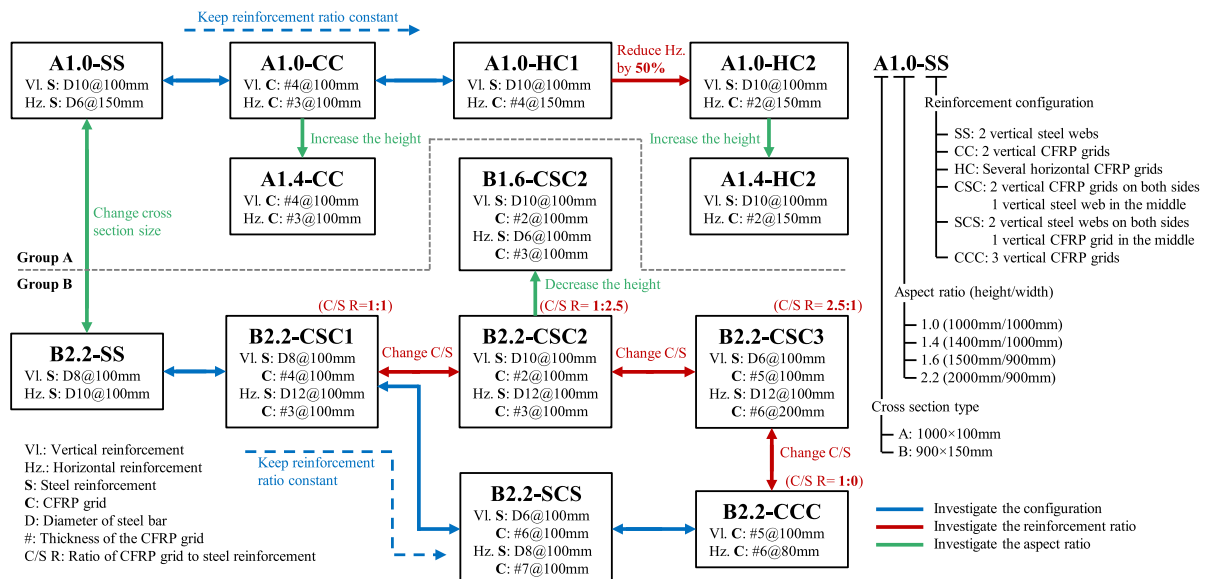


Fig. 1. Test program and specimen nomenclature.

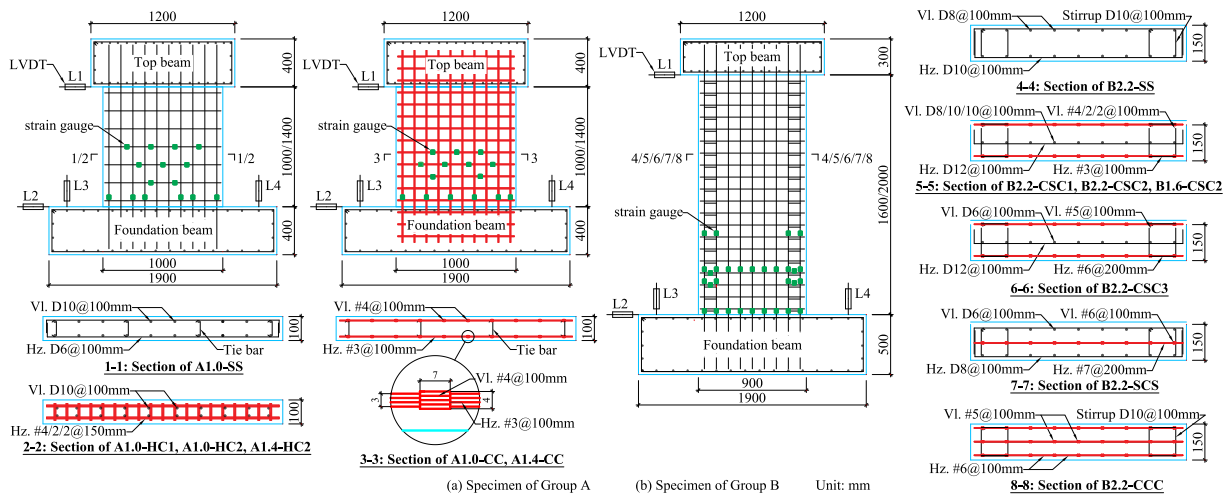


Fig. 2. Reinforcement details and instrumentation.

to 1400 mm. In Group B, specimen B2.2-SS was designed as a reference. Specimens B2.2-CSC1, B2.2-CSC2, and B2.2-CSC3 had the same total reinforcement ratio and configuration (two vertical CFRP grids on both sides and one steel web in the middle). The ratios of CFRP grid to steel bar of three specimens were different, which were 1:1, 1:2.5, and 2.5:1, respectively. Specimen B1.6-CSC2 was designed to reach a transition state, where the shear and flexural responses were both notable. Moreover, specimens B2.2-SCS and B2.2-CCC were added to consider the effect of reinforcement configuration, compared with specimens B2.2-SS and B2.2-CSC1. Specimen B2.2-SCS had two vertical steel webs on both sides and one CFRP grid in the middle. Similarly, specimen B2.2-CCC had three vertical CFRP grids as the reinforcement. To ensure the strength of edge grid nodes, the bars at the nodes extended outward for a certain length, as well as the certain thickness of concrete cover was considered.

Table 1 lists the reinforcement ratio of specimens. In Group A, the vertical reinforcement ratio was heavy set at 1.57% to avoid flexural failure, and the horizontal reinforcement was set at about 0.40% or 0.20%. Note that the vertical reinforcement ratio of specimens A1.0-CC and A1.4-CC was 0.56% due to the standardized production of CFRP grid. The lower reinforcement ratio was proved sufficient for flexural resistance in previous finite element simulation. Likewise, in Group B, the horizontal reinforcement ratio was heavy set at about 1.00% to

avoid shear failure (except for B1.6-CSC2) and the vertical reinforcement ratio was set at about 0.70%. Specimen B1.6-CSC2 was designed with a normal reinforcement ratio, functioning as a transition specimen. Designed parameters of specimens including the size and reinforcement ratio referred to Chinese standards GB50010-2010 [30] and GB50608-2010 [31]. Besides, ACI 440.1R-15 [32] and CSA S806-12 [33] were also taken as the reference for FRP material.

2.2. Material properties

Table 2 summarizes the tensile properties of single CFRP bar and steel bar. In this study, the CFRP grid was provided by Nanjing Loyalty Composite Equipment Manufacturing Co. LTD. The designed ultimate strength and elastic modulus were 2000 MPa and 120 GPa, respectively. Two types of CFRP grid, with a uniform length of 7 mm and thickness of 3 mm and 4 mm (named #3 and #4, see Fig. 2, Section 3-3), were tested according to Chinese standard GB/T 1446-2005 [34] and GB/T 36262-2018 [35]. The tensile test of steel bar was according to GB/T 228.1-2010 [36]. In addition, six concrete cube blocks (150 mm × 150 mm × 150 mm) and six concrete prismatic blocks (150 mm × 150 mm × 300 mm) were tested in each group according to GB/T 50081-2019 [37]. The cubic compressive strength and prism compressive strength in Group A were 31.8 MPa and 24.6 MPa, and in Group B were 40.8 MPa

Table 1
Reinforcement ratio of specimen.

Specimen	Vertical			Horizontal		
	Reinforcement	Ratio (%)	Total (%)	Reinforcement	Ratio (%)	Total (%)
A1.0-SS	D10@100	1.57	1.57	D6@150	0.40	0.40
A1.0-CC	#4@100	0.56	0.56	#3@100	0.42	0.42
A1.0-HC1	D10@100	1.57	1.57	#4@150	0.39	0.39
A1.0-HC2	D10@100	1.57	1.57	#2@150	0.20	0.20
A1.4-CC	#4@100	0.56	0.56	#3@100	0.42	0.42
A1.4-HC2	D10@100	1.57	1.57	#2@150	0.20	0.20
B1.6-CSC2	D10@100	0.52	0.71	D6@100	0.19	0.47
	#2@100	0.19		#3@100	0.28	
B2.2-SS	D8@100	0.67	0.67	D10@100	1.05	1.05
B2.2-CSC1	D8@100	0.34	0.71	D12@100	0.75	1.03
	#4@100	0.37		#3@100	0.28	
B2.2-CSC2	D10@100	0.52	0.71	D12@100	0.75	1.03
	#2@100	0.19		#3@100	0.28	
B2.2-CSC3	D6@100	0.19	0.66	D12@100	0.75	1.03
	#5@100	0.47		#6@200	0.28	
B2.2-SCS	D6@100	0.38	0.66	D8@100	0.67	1.00
	#6@100	0.28		#7@100	0.33	
B2.2-CCC	#5@100	0.70	0.70	#6@80	1.05	1.05

Notes: D means diameter of steel bar; # means thickness of CFRP grid; @ means spacing between reinforcements.

Table 2
Mechanical properties of CFRP grid and steel bar.

Material	A (mm ²)	f_y (MPa)	f_u (MPa)	ε_y (%)	ε_u (%)	E (GPa)
CFRP #3	21.73	–	2293	–	1.871	122.69
CFRP #4	28.15	–	2109	–	1.735	121.76
Steel D6	24.54	390	532	0.188	–	205.45
Steel D8	52.30	478	620	0.229	–	208.49
Steel D10	80.60	465	595	0.228	–	200.67
Steel D12	114.61	445	615	0.220	–	202.02

Notes: A means area of cross section; f_y means yield strength; f_u means ultimate strength; ε_y means yield strain; ε_u means ultimate strain; E means elastic modulus in tension; # means thickness of CFRP grid; D means diameter of steel bar.

and 29.5 MPa.

2.3. Test setup and loading program

Fig. 3 illustrates the test setup. The specimen was fixed on the rigid laboratory floor by four anchor bolts. Two lateral supports were set on each side of the specimen to avoid out-of-plane displacement. On top of the specimen, a loading beam was placed to distribute the axial load from the hydraulic jack above. The constant axial load, which was 10% of the concrete compressive strength, was applied through the hydraulic jack. A roller bearing was fixed on the reaction frame to provide the horizontal degree of freedom. A 100-ton MTS actuator was fixed on the reaction wall and connected to the lateral side of the top beam to provide the lateral load.

The loading history is shown in Fig. 4. In Group A, the displacement control loading was adopted (Fig. 4(a)). The displacement increased by 1 mm at each loading level. The specimen cycled twice in each level until its lateral load dropped to 85% of the ultimate capacity, where the specimen was considered as failure. In Group B, according to Chinese specification for seismic test of buildings (JGJ/T 101–2015) [38], the loading history of specimen with significant yield characteristics, included two stages: force control and displacement control. Before the vertical steel reinforcement yielded, the force control was adopted with one cycle in each level to find the yield displacement and the force increment of each cycle was 10 kN. After that, the displacement control was used with three cycles, where the increment of each level was named yield displacement (Δ_y). The specimen was regarded as failure when its load was less than 85% of the ultimate load.

2.4. Instrumentation

The lateral force and displacement were recorded by a transducer set in the MTS actuator. The data was transferred to a computer connected

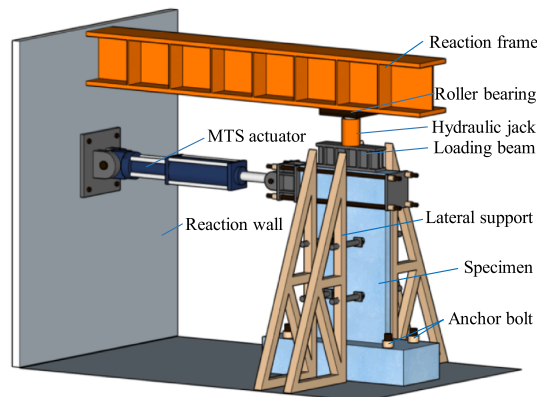


Fig. 3. Test setup.

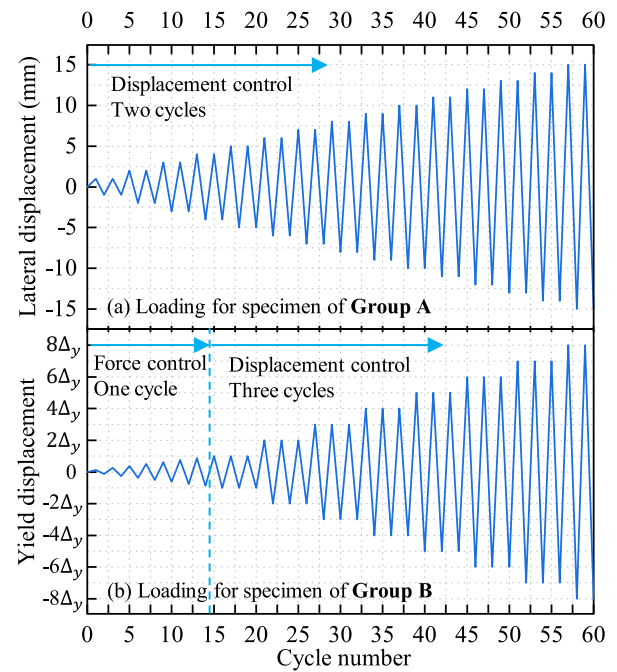


Fig. 4. Loading history.

to the transducer. The axial load was measured by a load cell set in the hydraulic jack on the loading beam. The linear variable differential transformers (LVDTs) and strain gauges were arranged to measure the deformation of the specimen and reinforcement, shown in Fig. 2. LVDT L1, mounted at the bottom of the top beam, was to check the lateral displacement. LVDT L2 was set on the top side of the foundation beam to monitor the sliding deformation. Two LVDTs, L3 and L4, were vertically fixed on the foundation beam to monitor warping. Moreover, strain gauges were set on the reinforcements to measure the strain development at different positions.

3. Results and discussion

3.1. Failure mode and crack propagation

Referring to the reinforced concrete shear wall, failure modes of CFRP grid shear wall were divided into shear failure, flexural failure, and the transition state, flexural-shear failure. Fig. 5 shows the crack propagation of specimens in different failure modes. The failure progress included three stages: initial, middle, and final stages. Table 3



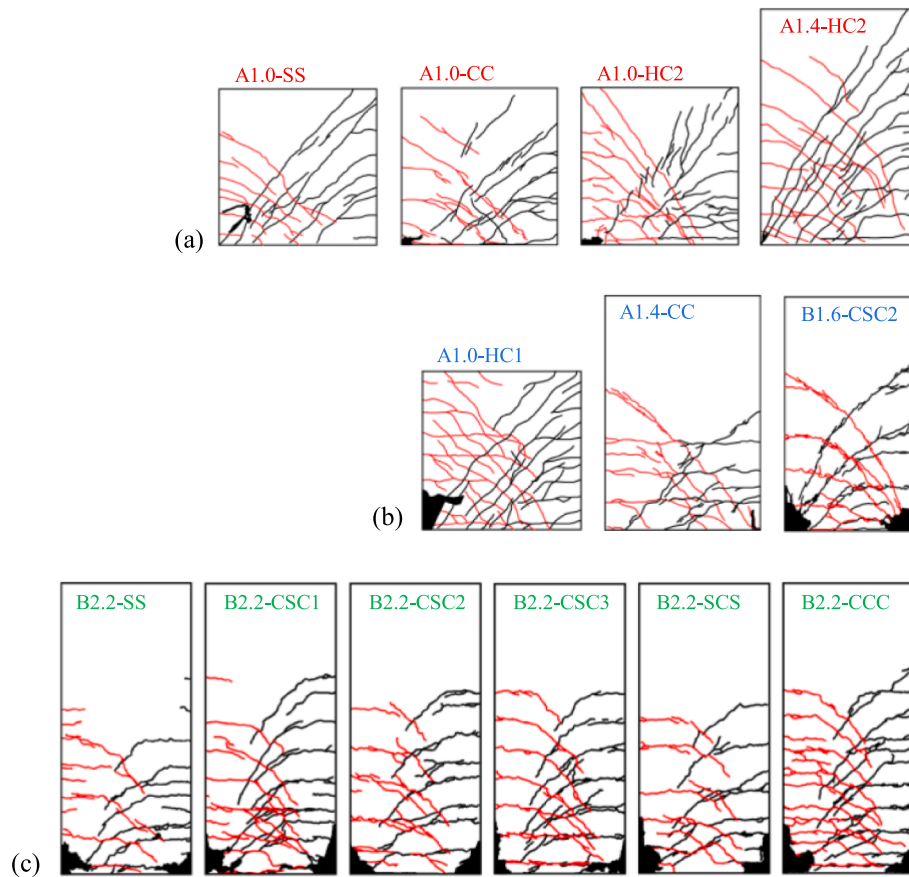


Fig. 5. Crack pattern of different failure modes: (a) shear failure; (b) flexural-shear failure; (c) flexural failure.

Table 3
Characteristics of different failure modes at three stages.

Failure mode	Initial stage	Middle stage	Final stage
Shear failure	Dia. C. developed Hz. C. developed (few)	Dia. C. developed and intersected Hz. C. tilted	Dia. C. developed and intersected Hz. S. yielded
Flexural-shear failure	Dia. C. developed (few) Hz. C. developed	Dia. C. developed Hz. C. connected (at bottom)	Dia. C. developed and intersected Hz. C. tilted Hz. S. and Vl. S. yielded
Flexural failure	Hz. C. developed Vl. S. yielded	Hz. C. connected Hz. C. tilted	Hz. C. tilted and intersected Vl. S. fractured/ Vl. CG. ruptured

Notes: Dia. C. means diagonal cracks; Hz. C. means Horizontal cracks; Hz. S. means horizontal steel bars; Vl. S. means vertical steel bars; Vl. CG. means vertical CFRP grids.

summarizes the characteristics of different failure modes at three stages. Fig. 6 illustrates the typical failure characteristics.

3.1.1. Shear failure

Specimens A1.0-SS, A1.0-CC, A1.0-HC2, and A1.4-HC2 showed typical shear failure (see Fig. 5(a)), where shear response dominated the whole loading process. In the initial stage, diagonal cracks appeared from two sides of the specimen, along with a few horizontal initial cracks. As the lateral displacement increased, to the middle stage, horizontal cracks immediately tilted to the diagonal direction, while diagonal cracks developed and intersected. In specimens A1.0-HC2 and A1.4-HC2, one horizontal crack connected through at the bottom

because the horizontal reinforcement ratio of these two specimens was half of that in other specimens, resulting in the flexural response at the bottom. In the final stage, vertical cracks gradually developed at the bottom (see Fig. 6(a)), indicating the specimen would reach the peak load. It found that several horizontal steel bars yielded, and the number of diagonal cracks increased.

As shown in Fig. 5(a), diagonal crack was the major crack pattern in shear failure. The width of the crack in CFRP grid specimen was lower than that in RC specimen, even when the horizontal reinforcement ratio of CFRP grid wall was cut in half. Moreover, a large number of hairlike cracks surrounded the main crack in CFRP grid specimen, which could also be observed in flexural-shear failure and flexural failure. It could be attributed to the property of elasticity of CFRP grid. Stress concentration was avoided in CFRP grid specimen, which meant a uniform strain distribution in CFRP grid, leading to the well-developed crack pattern and smaller width. On the contrary, in RC specimen, deformation concentration appeared where the steel bar yielded, resulting in a gradually larger crack width.

3.1.2. Flexural-shear failure

Specimens A1.0-HC1, A1.4-CC, and B1.6-CSC2 showed a transitional failure mode, flexural-shear failure, with the characteristics of both shear failure and flexural failure (see Fig. 5(b)). In the initial stage, horizontal cracks developed from two sides of the wall, while only a few shear cracks developed. It meant flexural response was dominant in the early stage, which was different from shear failure. When the load increased, to the middle stage, the horizontal cracks developed and got through, and several horizontal cracks tilted according to the diagonal. Meanwhile, diagonal cracks appeared not only from the side but in the middle of the wall, suggesting that shear response became dominant in the following process. In the final stage, diagonal cracks intersected and

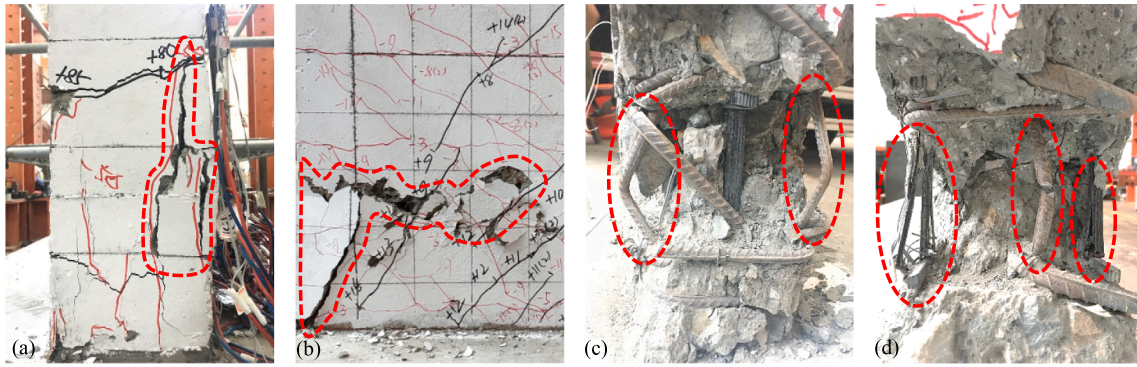


Fig. 6. Typical failure characteristics: (a) vertical compression cracks (side view); (b) concrete cover spalling (front view); (c) steel bars buckling (side view); (d) steel bar fracture and CFRP grid rupture (side view).

extended to the bottom. Concrete cover in the compression zone began to spall (see Fig. 6(b)). It also recorded that both vertical and horizontal steel bars yielded during this stage.

Fig. 5(b) shows a combined crack pattern with horizontal and diagonal cracks, indicating a tendency that flexural response was dominant in early stage, and then shear response was dominant in later stage. Compared to the RC specimen A1.0-SS, specimen A1.0-HC1 had the same reinforcement ratio but different reinforcement configuration. It showed specimen A1.0-HC1 with flexural-shear failure had finer crack and uniform crack propagation, as well as better load capacity and deformation performance. Specimen A1.4-CC shared the same reinforcement configuration with A1.0-CC, but larger aspect ratio, indicating that aspect ratio was an important factor in determining the failure mode of CFRP grid shear wall.

3.1.3. Flexural failure

Specimens B2.2-SS, B2.2-CSC1, B2.2-CSC2, B2.2-CSC3, B2.2-SCS, and B2.2-CCC showed a typical flexural failure (see Fig. 5(c)). In the initial stage, only horizontal cracks appeared on both sides. Vertical steel bars yielded when the lateral displacement was only 5 mm to 7 mm. After that, displacement control was applied. In the middle stage, horizontal cracks constantly appeared along the wall from bottom to top. Cracks at the bottom developed to the middle and connected, while those in the middle height tilted to the diagonal side. With the increased loading, to the final stage, horizontal cracks are titled and intersected. Under the large lateral displacement, vertical cracks appeared in compression zone, resulting in a large number of concrete cover spalling. Without the confinement of concrete, vertical steel bars buckled (see Fig. 6(c)) under compression and fractured under tension, while CFRP grids experienced brittle rupture (see Fig. 6(d)).

Fig. 5(c) shows the crack pattern of flexural failure. Compared to the RC specimen B2.2-SS, specimens B2.2-CSC1, B2.2-SCS, and B2.2-CCC, with the same reinforcement ratio, presented smaller crack width and uniform crack distribution, especially for specimen B2.2-CCC. This phenomenon was consistent with specimens in shear failure, which reflected the better deformation and carrying capacity of CFRP grid walls.

3.2. Hysteretic behavior

Fig. 7 shows the relationship between lateral load and displacement of specimens, where the lateral load and drift ratio of initial crack point (V_i, θ_i), yield point (V_y, θ_y), peak point (V_p, θ_p), and ultimate point (V_u, θ_u) are illustrated. Fig. 8 shows the load–displacement envelope curves of specimens, where the red line represents shear failure, the blue line represents flexural-shear failure, and the green line represents flexural failure. Table 4 summarizes the load and drift ratio of the characteristic point. In general, the hysteretic curve of the shear failure specimen presents a flatter loop with poor energy dissipation. Due to the lower

aspect ratio, specimens governed by shear response usually experienced a sudden failure with poor ductility but higher peak load. With the increase in aspect ratio, specimens of flexural-shear failure showed improved energy dissipation and deformation capacity. The hysteretic loop became plumper compared to that of shear failure specimen, however, poorer than that of flexural failure one. The perfect ductility and energy dissipation were achieved in flexural failure specimens, reflecting the plumper loops and larger displacement.

Compared to the RC specimens, CFRP grid specimens performed better hysteretic behavior, including load and deformation capacities in almost entire loading process. In Group A, the peak load of the wall with horizontally arranged CFRP grid (specimens A1.0-HC1 and A1.0-HC2) increased 10.4%–13.6% due to the confinement of CFRP grid to concrete. The grids functioned as stirrups, providing stronger confinement to concrete to enhance its compressive capacity, as a result, improving the load capacity. The lateral displacement of CFRP grid specimen increased 14.3%–78.6%, indicating an effective improvement of deformation capacity. It would be one of the advantages of low aspect ratio specimen to avoid brittle failure. Horizontally arranged CFRP grid specimen performed better deformation capacity compared to the vertically arranged one (specimen A1.0-CC), even the horizontal reinforcement ratio was only half (specimen A1.0-HC2). In Group B, the peak load showed an increase of 16.3%–33.2%, which was more significant compared to specimens in Group A. Due to the relative lower modulus and linear elasticity properties, CFRP grid can provide higher stress under the larger later deformation. It found that the lateral load of RC specimen B2.2-SS tended to be constant after the yield point. However, the load of specimen with CFRP grid showed continued growth, also named secondary stiffness. That was because steel bar yielded at large displacement, while CFRP grid could continuously provide high stress under large later deformation, resulting in the growing load. The lateral displacement of CFRP grid shear wall showed an increase of 8.0%–26.0%, which was lower than specimens in Group A. It was because the good ductility was one of the properties of high aspect ratio specimens. In general, CFRP grid could significantly improve the deformation capacity for lower aspect ratio specimens, while significantly enhancing the load capacity for higher aspect ratio specimens. It provided specific enhancement properties for specimens with different aspect ratios in further application.

3.3. Stiffness degradation

The secant stiffness K_i of specimens under cyclic load can be calculated by Eq. (1), where $F_i^+(F_i^-)$ is the positive (negative) maximum lateral load of the i^{th} loading level at the first cycle; $d_i^+(d_i^-)$ is the corresponding lateral displacement of $F_i^+(F_i^-)$.

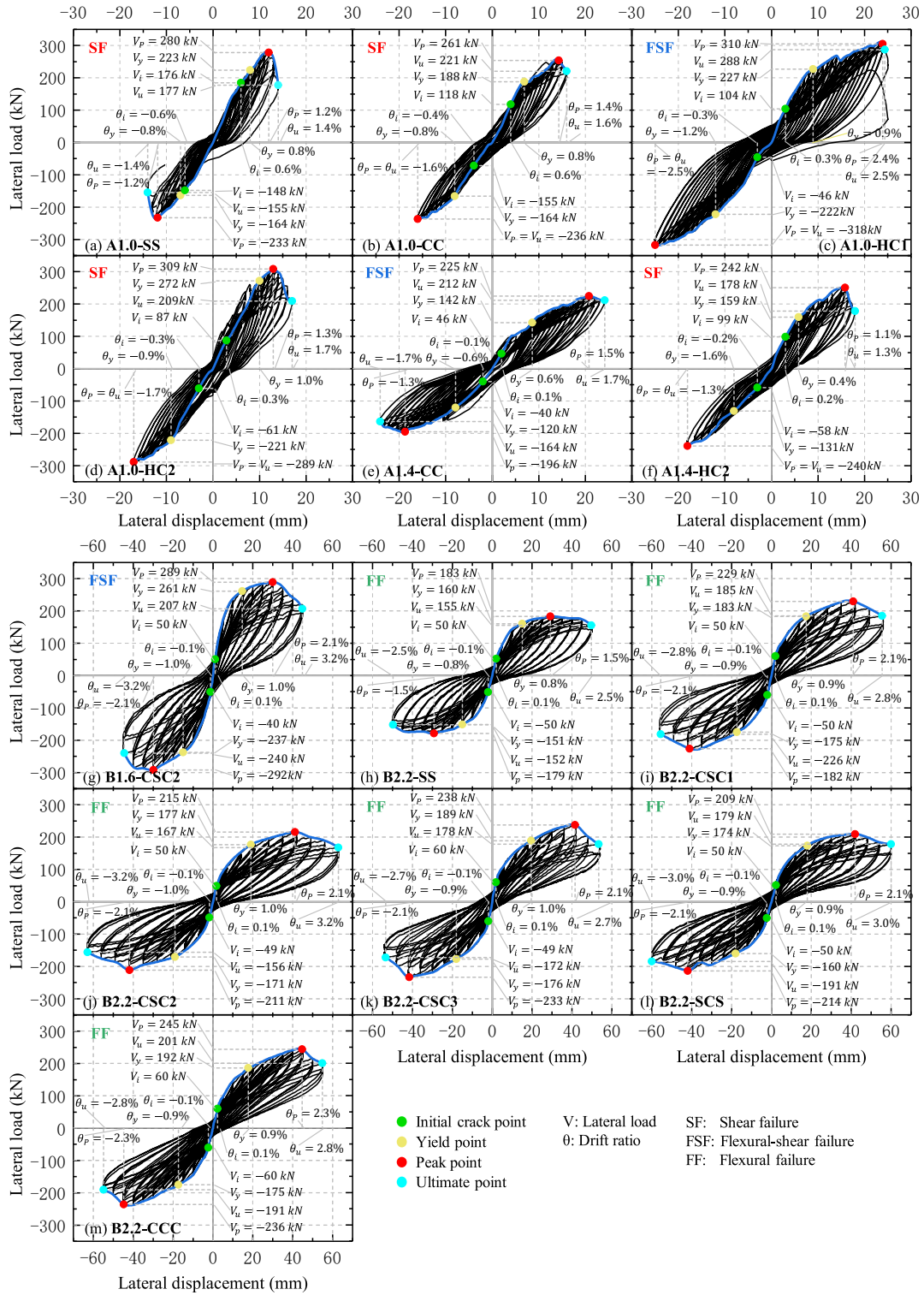


Fig. 7. Hysteretic curves.

$$K_i = \frac{|F_i^+| + |F_i^-|}{|d_i^+| + |d_i^-|} \quad (1)$$

Fig. 9 shows the scant stiffness degradation of specimens. Shear failure specimens had higher initial stiffness due to the lower aspect ratio, compared to the flexural-shear failure and flexural failure specimens. However, the stiffness degradation of shear failure specimens almost kept linear, while in flexural-shear failure and flexural failure

specimens, the rate of decline slowed with the increase of lateral displacement, reflecting a better ductility. Reinforced concrete specimen A1.0-SS showed a relatively higher initial stiffness, but it declined faster than specimens A1.0-CC and A1.0-HC1. It indicated more serious damage occurred in the RC specimen. In contrast, specimen with CFRP grid showed good confinement of concrete by grid, where specifically, the configuration of horizontally arranged grid was better than that of vertically arranged grid. Moreover, in flexural failure, the specimens

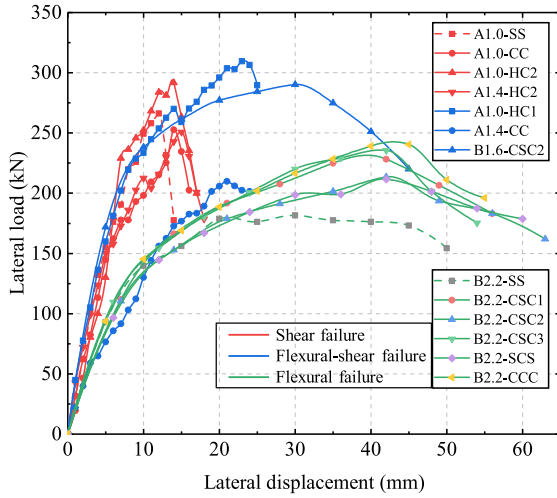


Fig. 8. Envelope curves.

shared almost the same stiffness in the initial stage. When the displacement was more than 20 mm, the stiffness of CFRP grid shear wall degraded more slowly, which exhibited the characteristic of secondary stiffness. The stiffness degradation was affected by the reinforcement configuration and CFRP grid proportion. As the CFRP grid was arranged outside, the damage would be effectively controlled before the rupture, leading to a slower stiffness degradation. Similarly, specimen with a higher proportion of CFRP grid would also achieve a slower rate.

3.4. Ductility

The ductility μ of specimens can be expressed as Eq. (2), where δ_u is the ultimate displacement of specimen, which is 85% of peak load in this study; δ_y is the displacement corresponding to the yield point.

$$\mu = \frac{\delta_u}{\delta_y} \quad (2)$$

Fig. 10 illustrates the ductility of specimens. In general, the ductility of specimens was positively related to the aspect ratio, which applied to both RC and CFRP grid shear walls. Actually, the ductility would be higher than the test result for the safety of experiment. The ultimate displacement δ_u was 85% of peak load in the test, however, more conservative than actual value. The ductility was effectively enhanced by 13.8%~36.7% by using CFRP grid in specimen with a lower aspect ratio (A1.0-CC and A1.0-HC1). Note that specimen A1.0-HC2 showed poorer ductility due to the half horizontal reinforcement ratio compared to the RC specimen, which experienced a premature failure with lower

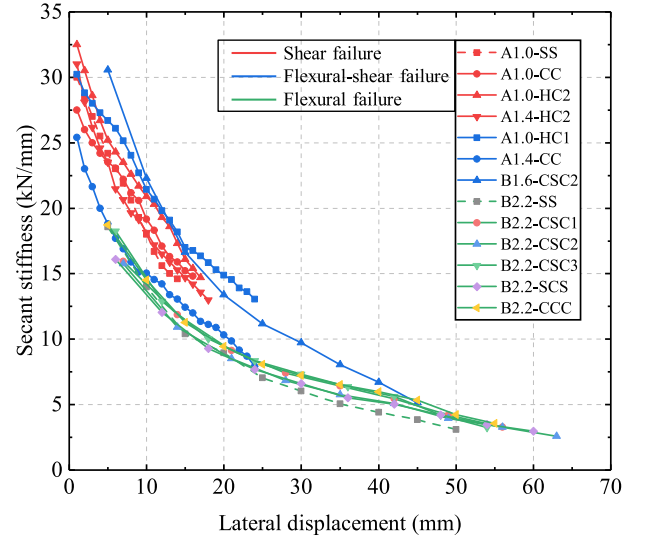


Fig. 9. Stiffness degradation.

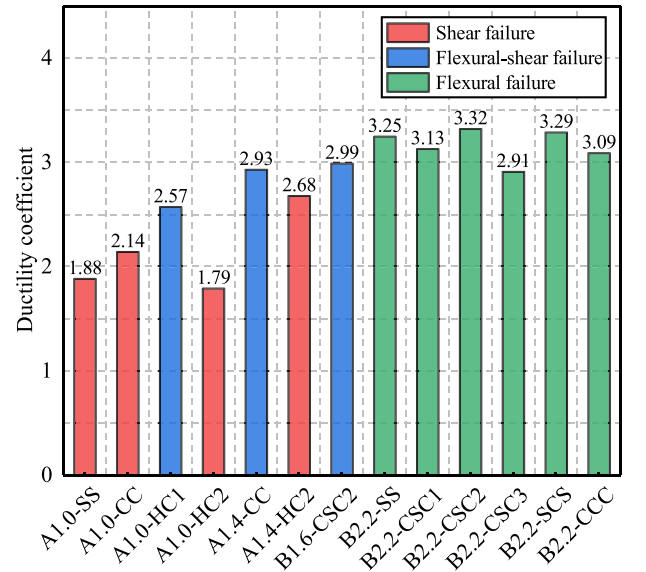


Fig. 10. Ductility coefficient.

Table 4
Summary of the characteristic points.

Specimen	Failure mode	V_i (kN)	θ_i (%)	V_y (kN)	θ_y (%)	V_p (kN)	θ_p (%)	V_u (kN)	θ_u (%)
A1.0-SS	SF	176	0.60	223	0.80	280	1.20	177	1.40
A1.0-CC	SF	118	0.60	188	0.80	261	1.40	221	1.60
A1.0-HC1	FSF	104	0.30	222	1.20	318	2.50	318	2.50
A1.0-HC2	SF	87	0.30	272	1.00	309	1.00	209	1.70
A1.4-CC	FSF	46	0.14	142	0.57	225	1.50	212	1.71
A1.4-HC2	SF	99	0.21	159	0.43	242	1.07	178	1.29
B1.6-CSC2	FSF	50	0.10	261	1.07	292	2.14	207	3.21
B2.2-SS	FF	50	0.11	160	0.75	183	1.50	155	2.50
B2.2-CSC1	FF	50	0.10	183	0.87	229	2.10	185	2.80
B2.2-CSC2	FF	50	0.10	177	0.95	215	2.10	167	3.15
B2.2-CSC3	FF	60	0.09	189	0.98	238	2.10	178	2.70
B2.2-SCS	FF	50	0.11	174	0.90	214	2.10	179	3.00
B2.2-CCC	FF	60	0.12	192	0.89	245	2.30	201	2.75

Notes: SF means shear failure; FSF means flexural-shear failure; FF means flexural failure.

ultimate displacement. The horizontal arranged CFRF grid of specimen A1.0-HC1 even improved the failure mode of the wall, resulting in an obvious increase in ductility. By contrast, the improvement of ductility of shear wall with grid in Group B was not prominent. It was because CFRP grid not only enhanced the deformation capacity of specimens, but delayed the yield point of specimens (see, Fig. 7 and Table 4, drift ratio of ultimate point, θ_u , and yield point θ_y). It showed the proportion of CFRP grid affected the ductility. In specimens B2.2-SS (0%), B2.2-CSC2 (25%), B2.2-CSC1 (50%), and B2.2-CSC3 (70%), the ductility of the specimen was optimized by the specific proportion. Specimen B2.2-SCS also performed better ductility because of the arrangement of outside steel webs and middle CFRP grid. The middle-laid grid would prolong the damage to internal concrete, inducing a higher ultimate displacement.

3.5. Energy dissipation

Fig. 11 illustrates the energy dissipation of each level and accumulation. The energy dissipation of each level can be calculated by the enclosed area in the hysteretic loop. The energy dissipation capacity depended on the lateral displacement and load capacity of specimen, which were related to failure mode. For flexural failure specimens, both each-level and accumulated energy dissipation were lower than that of flexural-shear failure and shear failure specimens at lower displacement. On one hand, specimens with shear and flexural-shear failure had a higher loading capacity under the same lateral displacement. On the other, the smaller displacement increment in each level (1 mm increment) was adopted in Group A, resulting in a more frequent energy accumulation. Although these specimens had higher energy dissipation, they experienced great damage early. In general, RC specimens performed better energy dissipation capacity compared to the CFRP grid specimen (see Fig. 11(a)), due to the different properties of steel bar and CFRP grid (plasticity and line elasticity). In addition, most CFRP grid specimens showed the same level of energy dissipation compared to RC specimen in Fig. 11(a). An exception of specimens B2.2-CCC and B2.2-CSC3 indicated that a high proportion of CFRP grid was not conducive to the energy dissipation of the specimen, which should be considered in structural design.

3.6. Residual deformation

Fig. 12 illustrates the residual deformation under different levels of lateral displacement, which relates to the self-centering performance of

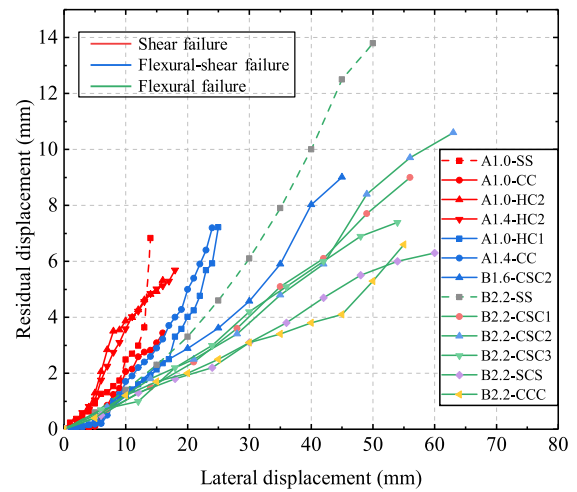


Fig. 12. Residual deformation.

the member. Before the appearance of the first crack, the residual displacement of all specimens was lower than 1 mm, where the specimens were in the elastic stage. It showed an obvious increase when horizontal and shear cracks developed. In general, CFRP grid specimens showed significantly smaller residual displacement (solid line in Fig. 12) compared to that of RC specimens (dash line in Fig. 12) with the same reinforcement ratio. The linear elasticity of CFRP grid provided the restoring stress to ensure the specimen back to the original state, resulting in a smaller residual deformation. While upon the steel bar yielded, the unrecoverable plastic deformation led to an obvious increasing residual deformation. Moreover, the residual deformation negatively correlated with the proportion of CFRP grid. The higher proportion of CFRP grid in specimens (25%, 50%, 70%, and 100% of specimens B2.2-CSC2, B2.2-CSC1, B2.2-CSC3, and B2.2-CCC), the lower residual displacement showed in Fig. 12. Note that the residual displacement of specimen B2.2-CCC experienced a rapid increase after the peak point due to the rupture of CFRP grid, which reflected the relatively lower ductility. The proper design of CFRP shear wall could take advantage of high strength and linear elasticity, to improve the crack propagation and reduced the residual deformation. As a result, the behavior of self-centering and less damage would reduce the risk of collapse and make it easy to be rehabilitated.

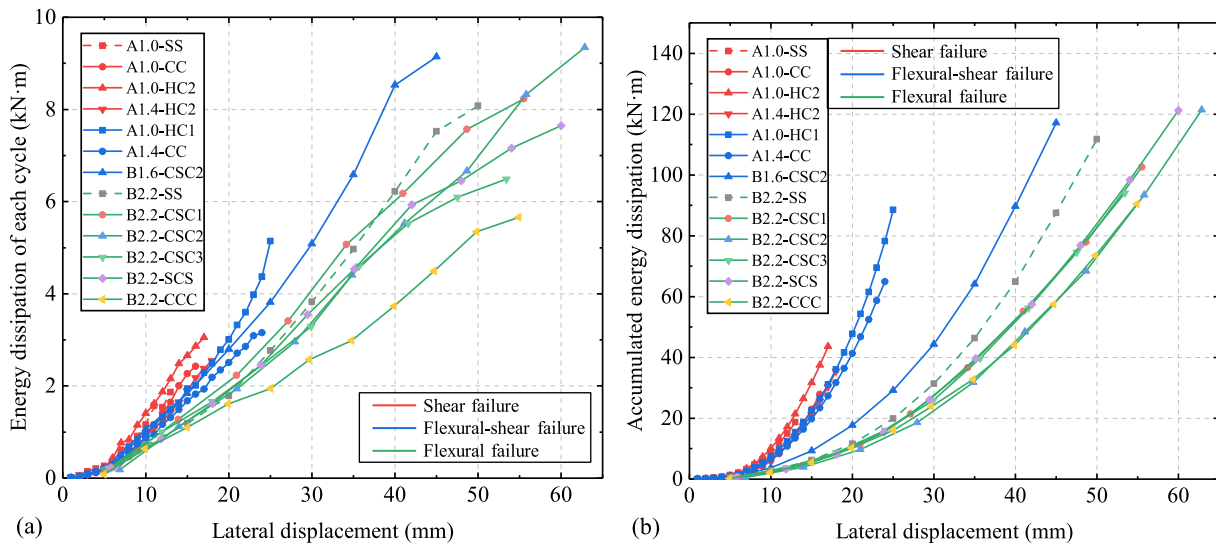


Fig. 11. Energy dissipation: (a) energy dissipation of each level; (b) accumulated energy dissipation.

3.7. Strain development

Strain development of reinforcement provided a micro perspective to investigate the mechanical property, as well as determine the failure mode of the specimen. The characteristics of reinforcement yielding under different failure modes are summarized in Table 3. Fig. 2 shows the arrangement of strain gauges. Representative gauges were selected to investigate the strain distribution of steel bar and CFRP grid.

3.7.1. Strain development of vertical reinforcement

Fig. 13 illustrates the typical strain distribution of vertical reinforcement at different drift ratios, noting that the horizontal axis is the distance to the centerline of the specimen. Gauges at the bottom of the specimen were selected to investigate the stress state of the bottom section.

In Group A, specimen A1.0-SS experienced shear failure, where the maximum strain of its vertical steel bar was 2131 $\mu\epsilon$, lower than the yield strain (2280 $\mu\epsilon$, see ϵ_y of steel D10 in Table 2). The strain distribution along the width of the specimen was linear during the load. By contrast, the strain of vertical steel bar in specimen A1.0-HC1 was 2127 $\mu\epsilon$, close to yielding strain at a drift ratio of 0.8%. In later loading, the vertical steel bar yielded, and its maximum strain reached 8000 $\mu\epsilon$ when the specimen failed. The difference on vertical strain distribution suggested that the horizontal arranged CFRP grid effectively increased the confinement to concrete, improved the stress state of specimen, and delayed the failure. As a result, specimen A1.0-HC1 experienced flexural-shear failure with better ductility. The tensile strain of CFRP

grid, however, became more than linear, especially for the boundary area of the reinforcement (specimens A1.0-CC and A1.4-CC). It was because the lower vertical reinforcement ratio of CFRP grid specimen and relatively lower elastic modulus induced a higher strain distribution. In specimen A1.4-CC, the maximum strain of CFRP grid was almost close to 18,000 $\mu\epsilon$, which was close to the ultimate tensile strain of the grid in the material experiment (see Table 2). It provided an example of how to take full advantage of high tensile strength for CFRP grid in design.

In Group B, the strain distribution of vertical reinforcement was almost linear, which proved the validity of the plane cross-section assumption. An exception of a drift ratio of 0.75% in specimen B2.2-SS could be explained as the failure of strain gauges (see Fig. 13(f)). The tensile strains of reinforcements experienced a large increase at the boundary area, maximums of 8000 $\mu\epsilon$ and 16,000 $\mu\epsilon$, for steel bar (specimen B2.2-CSC1) and CFRP grid (specimen B1.6-CSC2), respectively. Meanwhile, the interaction working behavior of concrete and CFRP grid also affected the cyclic performance of the specimen. Note that the maximum strain of CFRP grid in specimen with the aspect ratio of 2.2 was less than 14,000 $\mu\epsilon$ (specimen B2.2-SCS), only 75% of the ultimate strain. CFRP grid experienced rupture when the concrete in compressive zone cracked and spalled. Thus, concrete with higher compressive strength and ductility was suggested in design of CFRP grid shear wall with a high aspect ratio. In addition, in specimen B2.2-CCC, the strain of the inside CFRP grid was slightly higher than that of the outside CFRP grid. CFRP grid in the middle of the wall, wrapped in core concrete, experienced less damage compared to the outside, performing

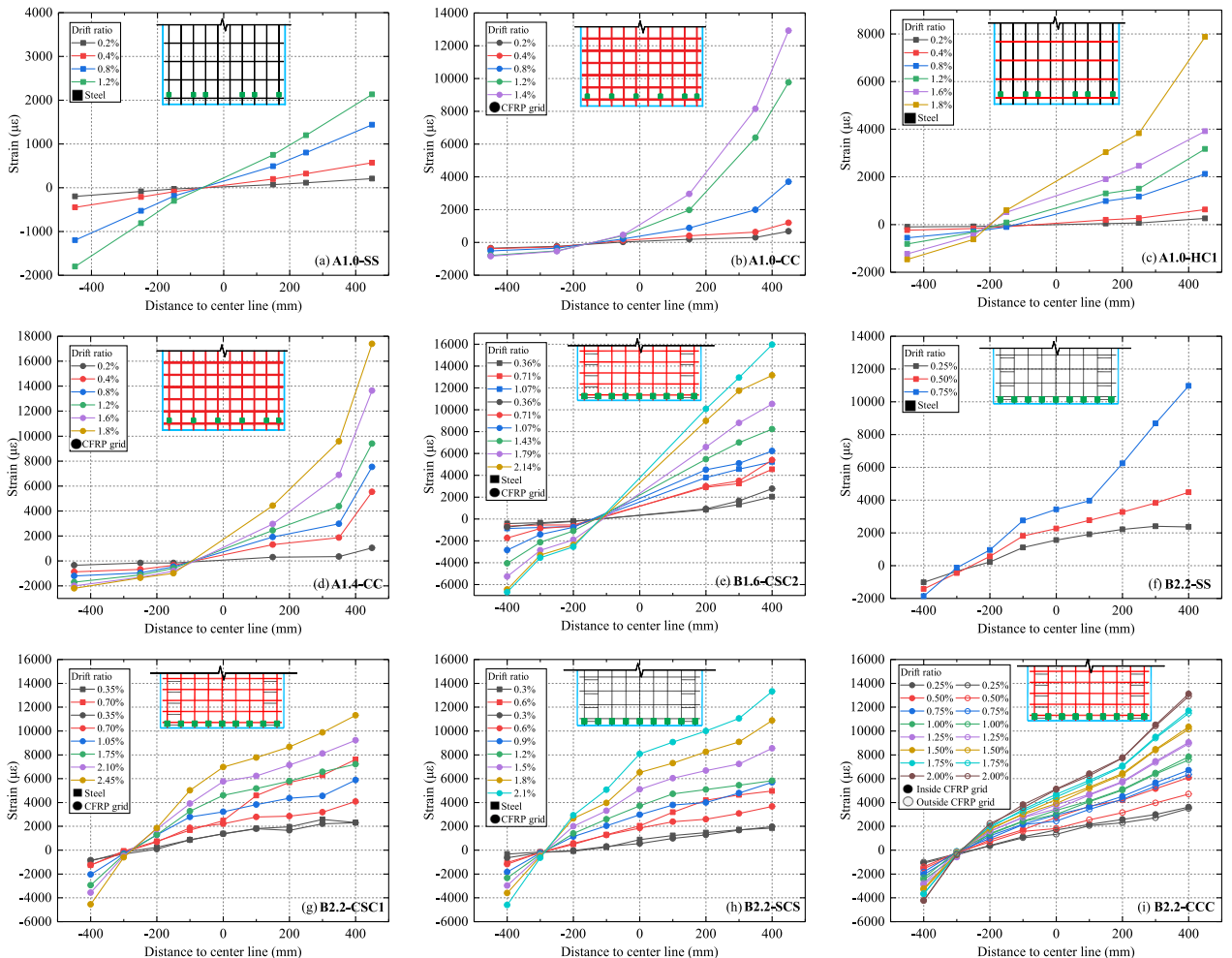


Fig. 13. Strain distribution of vertical reinforcement based on the distance to center line.

better deformation capability. This also revealed the importance of the interaction working behavior of concrete and CFRP grid.

3.7.2. Strain development of horizontal reinforcement

Fig. 14 illustrates the typical strain distribution of horizontal reinforcement, noting that the horizontal axis is the lateral displacement. Gauges in the area with significant shear response were selected to investigate the shear behavior of the specimen. The horizontal strain of Group B was not provided because the strain of the measuring point was lower and relatively stable. It was consistent with the failure progression in a high aspect ratio specimen with a weak shear response.

In Group A, all the selected horizontal steel bars in specimen A1.0-SS yielded in the final stage, with a maximum strain of $4000 \mu\epsilon$ during failure, showing a dominant shear response in the failure progression. In CFRP grid specimens, the average maximum strains of the selected gauges were more than $4000 \mu\epsilon$ in the horizontal arranged grid (specimens A1.0-HC1, A1.0-HC2, and A1.4-HC2), and $3000 \mu\epsilon$ in the vertical arranged grid (specimens A1.0-CC and A1.4-CC), which was only $2500 \mu\epsilon$ in RC specimen. On one hand, the relative lower elastic modulus partial led to the higher strain. On the other, the good integrity of CFRP grid enhanced the deformation ability of the core concrete, which in turn increased the strain of the grid. Especially for specimens A1.0-HC1, A1.0-HC2 and A1.4-HC2, the horizontal arranged CFRP grid worked as the stirrups to provide strong confinement for concrete, resulting in increases in loading capacity and ductility. When the horizontal reinforcement was reduced to half, specimen A1.0-HC2 showed only a little decrease in load capacity, but a significant reduction in ductility.

4. Conclusion

To investigate the cyclic behavior of concrete shear wall with CFRP grid-steel reinforcement, thirteen pieces of specimens were tested. Considering the parameters of reinforcement ratio, reinforcement configuration, and aspect ratio, the mechanical property of shear was analyzed based on the hysteretic behavior, stiffness degradation, ductility, energy dissipation, residual deformation, and strain development. The conclusions are drawn as follows:

- (1) Compared to the RC wall, CFRP grid shear wall showed smaller width of main cracks, surrounded by large number of hairlike cracks. Diagonal cracks accounted for most shear failure specimens, while horizontal cracks accounted for most flexural failure specimens, which was corresponding with the RC shear wall.
- (2) CFRP grid specimens performed better hysteretic behavior. For specific, CFRP grid could significantly improve the deformation capacity for the lower aspect ratio specimen, while significantly enhancing the load capacity for the higher aspect ratio specimen. It provided specific enhancement properties for specimens with different aspect ratios in further application.
- (3) The stiffness degradation of shear failure specimens almost kept linear, while in flexural-shear failure and flexural failure specimens, the rate of decline slowed with the increase of lateral displacement, reflecting a better ductility. Compared to the RC wall, the stiffness of CFRP grid specimens with high aspect ratio degraded more slowly, exhibiting the characteristic of secondary stiffness.
- (4) The ductility of specimens was positively related to the aspect ratio. In specimen with lower aspect ratio, the ductility was effectively enhanced by 13.8%~36.7% by using CFRP grid. In specimen with higher aspect ratio, the improvement of ductility of CFRP grid specimen was not prominent because CFRP grid not only enhanced the deformation capacity of specimens, but delayed the yield point of specimens.
- (5) RC specimens performed better energy dissipation capacity compared to the CFRP grid specimen due to the different properties of steel bar and CFRP grid (plasticity and line elasticity). However, most CFRP grid specimens showed the same level of energy dissipation compared to RC specimen through a reasonable design.
- (6) CFRP grid specimens showed significantly smaller residual displacement compared to that of RC specimens with the same reinforcement ratio, indicating the characteristics of self-centering. Taking advantages of the high strength and linear elasticity of CFRP grid would prevent structural collapse and reduce the damage.
- (7) The strain of reinforcement in CFRP grid specimens was increased compared to RC wall due to the linear elasticity of the grid and its

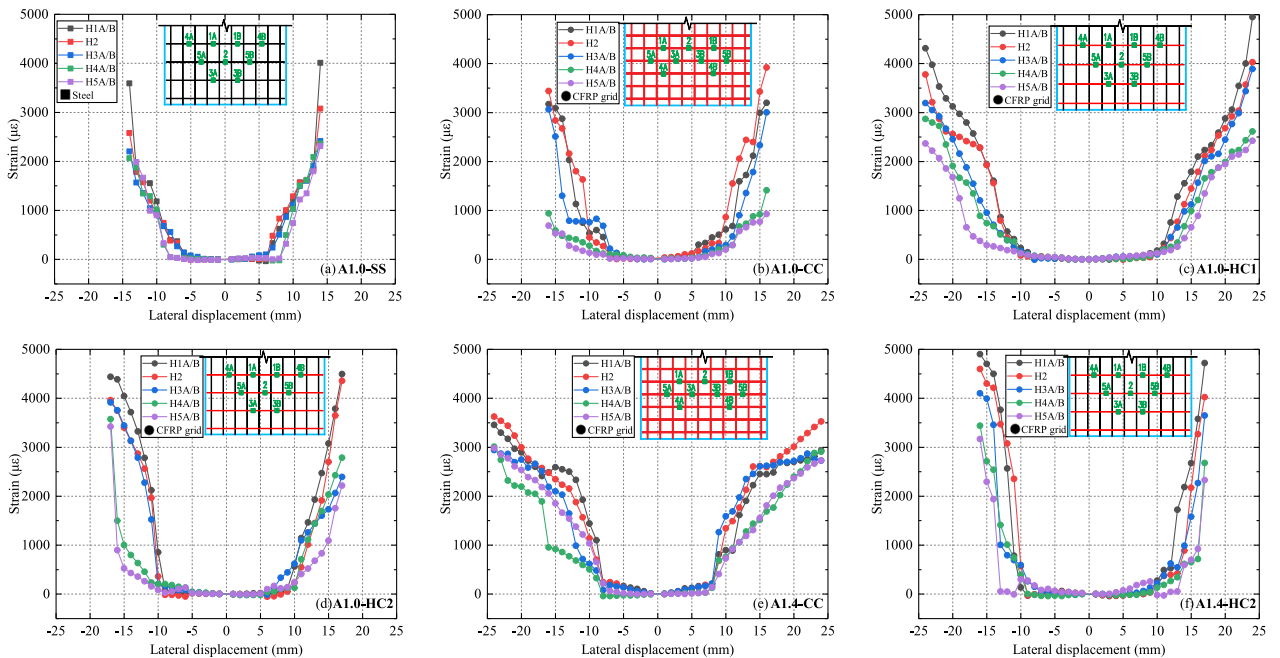


Fig. 14. Strain distribution of horizontal reinforcement based on lateral displacement.

confinement to concrete. To take full advantage of the high strength of CFRP grid, concrete with higher compressive resistance and ductility was suggested to use in compression zone.

CRedit authorship contribution statement

Jie Shen: Validation, Investigation, Writing – original draft, Writing – review & editing, Visualization, Data curation. **Zhen Huang:** Conceptualization, Writing – review & editing, Supervision, Project administration. **Xinyu Song:** Methodology, Investigation, Writing – review & editing. **Huaquan Lin:** Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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